

DISASTER MANAGEMENT ENGINE ASSAM REGION
PROGRAM OUTPUT & FUNCTION TEST REPORT

Overall Result: PASS
Primary Algorithm: Dijkstra's Shortest Path
Storage: Binary File – relief_camps.dat

REGISTER RELIEF CAMP DATA

Purpose: Registers a new relief camp and saves its information into binary storage.

Camp ID	: 101
Camp Name	: Camp Alpha
Maximum Capacity	: 100
Current Occupancy	: 80
Food Stock	: 150 Kg
Medical Kits	: 10

[SUCCESS] Camp details saved to binary storage ('relief_camps.dat').

STATUS : PASS

DISPLAY ALL CAMPS

Purpose: Displays all registered relief camps along with occupancy, food stock, and medical kits.

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ID	Camp Name	Occupancy	Food (Kg)	Med Kits
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101	Camp Alpha	80/100	150	10
102	Camp Beta	50/50	75	5
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STATUS : PASS

SAFE EVACUATION ROUTE – DIJKSTRA

Purpose: Finds the shortest safe evacuation route while avoiding roads where flood-water depth exceeds 0.5 metres.

Guwahati Hub → Morigaon	50 km	Water: 0.2 m	PASSABLE
Morigaon → Nagaon	70 km	Water: 0.8 m	BLOCKED
Guwahati Hub → Nagaon	110 km	Water: 0.1 m	PASSABLE
Nagaon → Jorhat	90 km	Water: 0.3 m	PASSABLE
Jorhat → Majuli Shelter	20 km	Water: 0.1 m	PASSABLE

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OPTIMAL SAFE EVACUATION ROUTE (FLOOD-AWARE)

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Start Location : Guwahati Hub
Destination : Majuli Shelter
Total Distance : 220 KM
Path Taken : Guwahati Hub -> Nagaon -> Jorhat -> Majuli Shelter

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Distance Calculation: 110 km + 90 km + 20 km = 220 km
STATUS : PASS | ALGORITHM : DIJKSTRA | SAFE PATH : 220 KM

RESOURCE SHORTAGE ALERTS

Purpose: Checks all relief camps for food stock below 100 Kg and occupancy at or above maximum capacity

Camp Alpha – Normal Condition

Food Stock : 150 Kg Occupancy : 80/100

[STATUS] All camps operating with sufficient stocks and safe capacities.

STATUS : SAFE

Camp Beta – Critical Condition

Food Stock : 75 Kg Occupancy : 50/50

[ALERT] Camp ID: 102 (Camp Beta)

-> CRITICAL FOOD SHORTAGE: 75 Kg remaining!

-> CAMP OVERCROWDED: 50/50 seats filled!

STATUS : PASS

MAIN CONTROL LOOP

Purpose: Controls the complete program through the main menu.

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DISASTER MANAGEMENT ENGINE (ASSAM REGION)

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Register Relief Camp Data

View All Camps & Occupancy

Find Safe Evacuation Route (Dijkstra)

Scan Offline Resource Shortage Alerts

Exit System

Select Choice [1-5]:

Invalid Menu Choice Test

Input : 9

Output : Invalid choice. Select between 1 and 5.

STATUS : PASS

Invalid Text Input Test

Input : abc

Output : Invalid input. Please enter a number from 1 to 5.

STATUS : PASS

Exit Test

Input : 5

Output : Exiting System. Stay Safe!

STATUS : PASS

FINAL TEST SUMMARY

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FINAL FUNCTION TEST REPORT

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registerCamp()

Test : Camp data registration

Result : PASS

displayCamps()

Test : Display stored camp records

Result : PASS

calculateEvacuationRoute()

Test : Flood-aware Dijkstra routing

Result : PASS

checkResourceAlerts()

Test : Food and occupancy monitoring
Result : PASS
main()
Test : Menu, invalid input and exit
Result : PASS

TOTAL FUNCTIONS TESTED : 5
FUNCTIONS PASSED : 5
FUNCTIONS FAILED : 0
OVERALL RESULT : PASS

SAFE EVACUATION ROUTE:

Guwahati Hub

↓

Nagaon

↓

Jorhat

↓

Majuli Shelter

TOTAL DISTANCE : 220 KM

BLOCKED ROAD:

Morigaon → Nagaon Flood Depth : 0.8 m

CRITICAL CAMP:

Camp Beta

Food Stock : 75 Kg

Occupancy : 50/50

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SYSTEM TEST COMPLETED

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